



4K @ 60Hz on Linux

Fix guide for Virtual Pinball cabinets

Bazzite / Nobara · Nvidia GPU · Samsung TV

The Problem

You built an awesome virtual pinball cabinet running Linux (Bazzite or Nobara), your Nvidia GPU is connected to a Samsung 4K TV via HDMI, and:

- Linux only offers 1920x1080 @ 60Hz over HDMI
- If you manually force 4K, the screen says Out of Range
- Using an HDMI-to-DisplayPort adapter gives you 4K but only at 30Hz
- Everything works fine in Windows — so the hardware is not broken

i NOTE This is a very common issue in the pinball community and it has a simple fix. The root cause is almost always the Samsung TV itself, not Linux.

Why Does This Happen?

Samsung TVs ship with a setting called HDMI UHD Color that is OFF by default. When it is off, the TV advertises itself to every connected device — including your GPU — as an HDMI 1.4 device. HDMI 1.4 has a bandwidth ceiling that caps 4K at 30Hz and prevents 60Hz entirely.

Windows quietly works around this by using a compressed color format (YCbCr 4:2:0) without telling you. The image looks okay but colour quality is reduced. Linux takes the TV at its word and simply refuses to offer modes the hardware claims it cannot do.

! IMPORTANT The fix is to tell the Samsung TV to stop lying about its capabilities. Once HDMI UHD Color is enabled, the TV correctly advertises HDMI 2.0 support and 4K @ 60Hz becomes available immediately.

Step-by-Step Fix

Step 1 — Enable HDMI UHD Color on your Samsung TV

Using your Samsung TV remote, navigate to:

[Settings](#) → [General](#) → [External Device Manager](#) → [HDMI UHD Color](#)

Set it to ON for the HDMI port your PC is connected to (usually HDMI 1 or HDMI 2).

i NOTE After turning this on, your TV may go black for a second and reconnect. That is normal — it is re-negotiating the signal with your GPU.

Now go to your Linux display settings (or right-click desktop → Display Settings) and check if 3840x2160 @ 60Hz is now available. For many people this single step solves everything.

✓ SUCCESS If 4K @ 60Hz now appears — you are done! The steps below are optional but recommended to make the fix permanent and to improve driver behaviour.

Step 2 — Apply the Nvidia HDMI 2.0 Driver Fix (Permanent)

Even with HDMI UHD Color enabled, the Nvidia driver on Linux sometimes still under-negotiates the HDMI version after a reboot. This one-liner permanently fixes that:

```
echo 'options nvidia NVreg_UseHDMI2deepColor=1' | sudo tee /etc/modprobe.d/nvidia-hdmi.conf
```

Verify it was written correctly:

```
cat /etc/modprobe.d/nvidia-hdmi.conf

# Should output exactly one line:
# options nvidia NVreg_UseHDMI2deepColor=1
```

Then rebuild your initramfs so the setting is loaded at boot:

```
sudo dracut --force
```

i NOTE On standard Fedora-based systems use: `sudo dracut --force` On Debian/Ubuntu-based systems use: `sudo update-initramfs -u`

Now reboot:

```
reboot
```

Step 3 — Verify After Reboot

After rebooting, open a terminal and run:

```
xrandr
```

Look for your Samsung display (usually HDMI-A-1). You should now see 3840x2160 listed with 60Hz available. The currently active mode will have an asterisk (*) next to it.

If it is not yet set to 4K @ 60Hz, open your display settings and select it manually. It should stick from now on.

Diagnostic Commands (Advanced)

If the steps above did not work, these commands will help diagnose exactly what is going on.

Check what your TV is advertising to Linux

```
sudo cat /sys/class/drm/card1-HDMI-A-1/edid | edid-decode

# Install edid-decode if needed:
sudo dnf install edid-decode      # Nobara / Fedora
sudo rpm-ostree install edid-decode  # Bazzite (then reboot)
```

Look for the line: Maximum TMDS clock. If it says 300 MHz, the TV is still advertising HDMI 1.4. Go back and make sure HDMI UHD Color is ON. After enabling it, the value should jump to 600 MHz.

Check what resolutions and refresh rates are available

```
xrandr
```

Your 4K display should appear as HDMI-A-1. Look for 3840x2160 entries. If all refresh rates are 29.xx Hz, the TV is still in HDMI 1.4 mode.

Check which Nvidia driver is loaded

```
nvidia-smi  
lsmod | grep nvidia
```

Make sure the proprietary Nvidia driver is loaded, not the open-source Nouveau driver. Bazzite and Nobara install the proprietary driver by default, but it is worth confirming.

Frequently Asked Questions

Why did Windows work but Linux didn't?

Windows silently uses a compressed colour format (YCbCr 4:2:0) to squeeze 4K @ 60Hz through an HDMI 1.4 connection. It never tells you it is doing this. Linux refused to do that without explicit permission, which is actually more correct behaviour. After this fix, Linux runs at true 4K @ 60Hz in full colour.

Why does the HDMI-to-DisplayPort adapter only give 30Hz?

Passive HDMI-to-DP adapters do not convert the signal — they just reshape the plug. The bandwidth is still limited by whatever HDMI version the TV is advertising. With HDMI UHD Color off, that is HDMI 1.4 which caps 4K at 30Hz. The best long-term solution is to connect directly via DisplayPort if your TV has a DP input.

My Samsung menu looks different — I can't find HDMI UHD Color

On newer Samsung TVs and monitors the setting may be called Input Signal Plus or HDMI Black Level. Check under Connection, External Device Manager, or Input settings depending on your model year. The option to look for is anything that enables HDMI 2.0 or higher bandwidth on a specific port.

Will this fix survive updates?

Yes. The modprobe config file in `/etc/modprobe.d/` persists across system updates. On Bazzite (which uses rpm-ostree), the file is in the writable overlay and will survive OS image updates.

Guide written with help from the community · Tested on Bazzite with Nvidia RTX 3070

Share freely — the more cabinets running at 4K @ 60Hz the better 🍷